One-hot XOR reduction → indexed warp shuffle

NVIDIA GB300 · SM 10.3 · 18 matched input sizes · 1.25× geometric-mean speedup

One Triton program, compiled twice

Only the compiler optimization changes

```
@triton.jit(do_not_specialize=["N", "seed"])
def _classic_one_hot_xor_kernel(indices_ptr, output_ptr, N, seed,
    BLOCK: tl.constexpr, BENCHMARK_VARIANT: tl.constexpr):
    offsets = tl.program_id(0) * BLOCK + tl.arange(0, BLOCK)
    mask = offsets < N
    index = tl.load(indices_ptr + offsets, mask=mask, other=0).to(tl.uint32)
    bits = tl.arange(0, 32)
    bases = (bits.to(tl.uint32) * 0x9E3779B9) ^ seed.to(tl.uint32)

    result = tl.full(index.shape, 0, tl.uint32)
    for bit in tl.static_range(32):
        basis = tl.xor_sum(tl.where(bits == bit, bases, 0), axis=0)
        result ^= tl.where((index >> bit) & 1, basis, 0)

    tl.store(output_ptr + offsets, result, mask=mask)
```

Bases are calculated

32 values stay in registers

No basis pointer or HBM basis load

optimization OFF

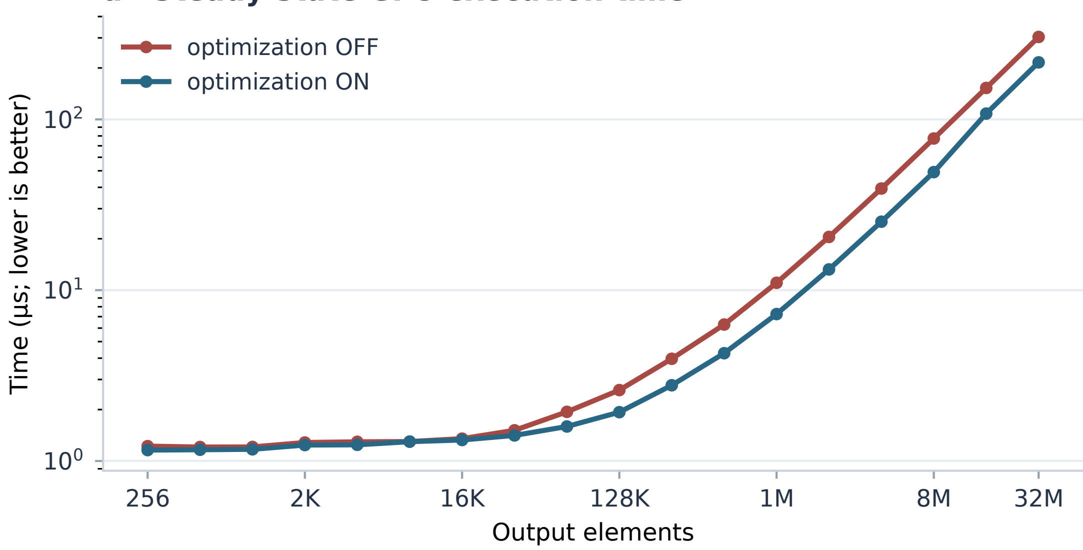
32 warp reductions

optimization ON

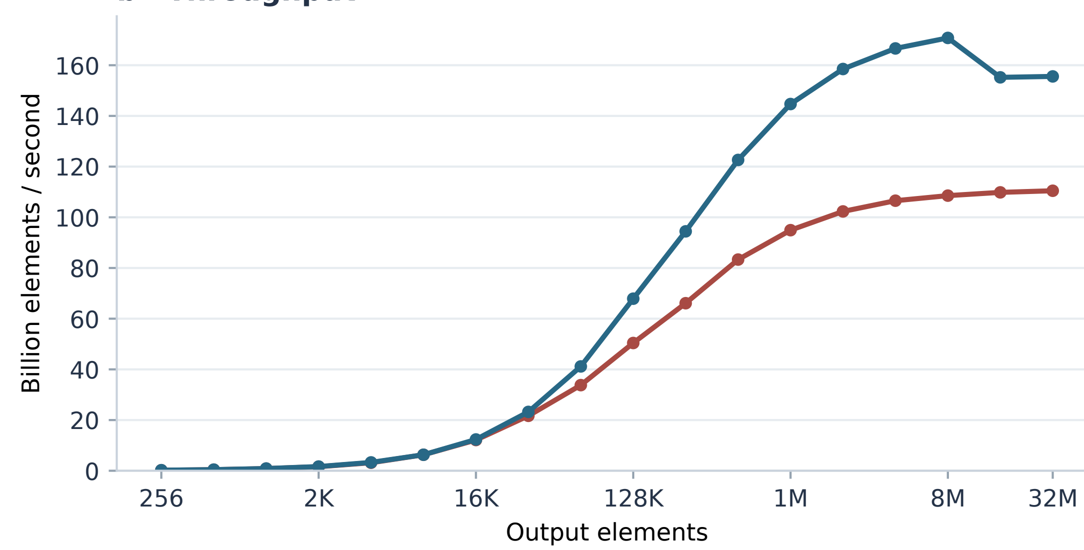
32 indexed shuffles

Same source, inputs, and GPU

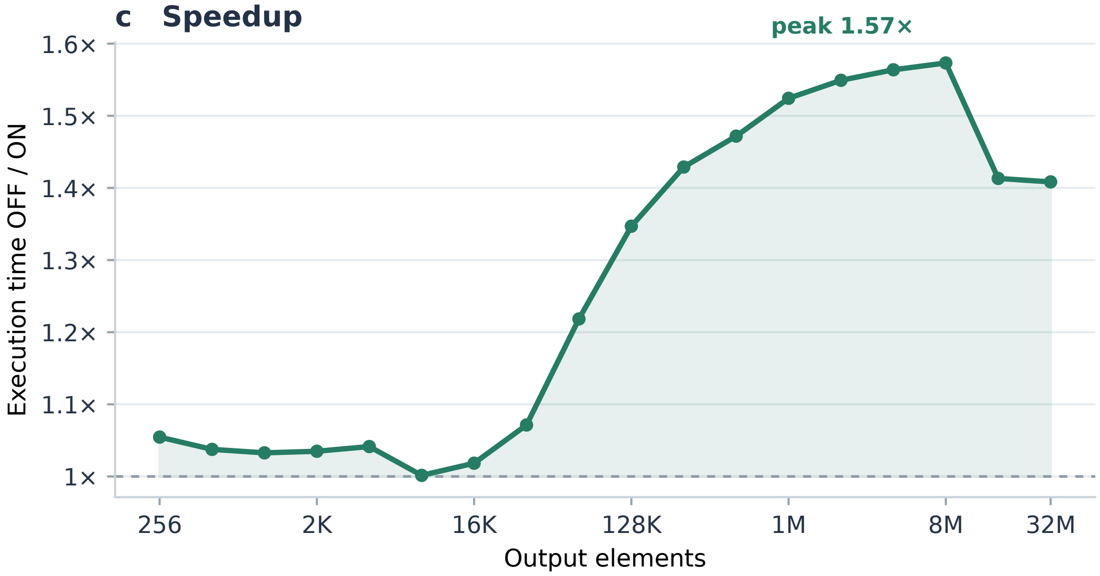
a Steady-state GPU execution time



b Throughput



c Speedup



d Generated code

optimization OFF

`redux.sync.xor.b32`

optimization ON

`shfl.sync.idx.b32`
`lop3.b32`

Static PTX / resource	OFF		ON
Total instructions	212	→	**117**
Warp-wide reductions	32	→	**0**
Indexed shuffles	0	→	**32**
Fused XOR operations	0	→	**31**
Registers per thread	30	→	**20**
Global loads (index only)	1	→	**1**
Basis HBM loads	0	→	**0**

CUDA Graph replay: reused input/output buffers, no L2 flush; median with 20th–80th percentile shading.
Warm-cache timing where the working set fits; larger working sets may still access HBM.